Newman-Penrose Formalism Calculations

Stephani et al., Equation (36.24)

Metric

$$ds^2 = -\mathrm{d}t \otimes \mathrm{d}t - (F+G)a\mathrm{d}t \otimes \mathrm{d}x - (F+G)a\mathrm{d}x \otimes \mathrm{d}t + \left(-\left(F^2 + 2\,FG + G^2\right)a^2 + \left(F^2 - 2\,FG + G^2\right)ab + \left(\frac{\partial\,F}{\partial t}\right)^2 + 2\,\frac{\partial\,F}{\partial t}\frac{\partial\,G}{\partial t} + \frac{\partial\,G}{\partial t}^2\right)\mathrm{d}x \otimes \mathrm{d}x + (F-G)\sqrt{a}\sqrt{b}\mathrm{d}x \otimes \mathrm{d}y + (F-G)\sqrt{a}\sqrt{b}\mathrm{d}y \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y + \mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$(x^\mu) = (t, x, y, z)$

Metric Functions

$a(x) = a$

$b(x) = b$

$F(t, x, y) = F$

$G(t, x, y) = G$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\right)\,dt + \left(\frac{1}{2}\sqrt{2}Fa + \frac{1}{2}\sqrt{2}Ga\right)\,dx + \left(\frac{1}{2}\sqrt{2}\right)\,dz$$

$$n = \left(\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{1}{2}\sqrt{2}Fa + \frac{1}{2}\sqrt{2}Ga\right) dx + \left(-\frac{1}{2}\sqrt{2}\right) dz$$

$$m = \left(\frac{1}{2}i\sqrt{2}\sqrt{ab}F - \frac{1}{2}i\sqrt{2}\sqrt{ab}G + \frac{1}{2}\sqrt{2}\frac{\partial}{\partial t}F + \frac{1}{2}\sqrt{2}\frac{\partial}{\partial t}G\right) dx + \left(\frac{1}{2}i\sqrt{2}\right) dy$$

$$\overline{m} = \left(-\frac{1}{2}i\sqrt{2}\sqrt{ab}F + \frac{1}{2}i\sqrt{2}\sqrt{ab}G + \frac{1}{2}\sqrt{2}\frac{\partial}{\partial t}F + \frac{1}{2}\sqrt{2}\frac{\partial}{\partial t}G\right) dx + \left(-\frac{1}{2}i\sqrt{2}\right) dy$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{1}{2}\sqrt{2}\right) dz$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(-\frac{1}{2}\sqrt{2}\right) dz$$

$$m = \left(-\frac{\sqrt{2}Fa}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}Ga}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}\right) dt + \left(\frac{1}{\sqrt{2}\frac{\partial}{\partial t}F + \sqrt{2}\frac{\partial}{\partial t}G}\right) dx + \left(-\frac{\sqrt{2}F\sqrt{a}\sqrt{b}}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}G\sqrt{a}\sqrt{b}}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\frac{\partial}{\partial t}F}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}\right) dy$$

$$\overline{m} = \left(-\frac{\sqrt{2}Fa}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}Ga}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}\right) dt + \left(\frac{1}{\sqrt{2}\frac{\partial}{\partial t}F + \sqrt{2}\frac{\partial}{\partial t}G}\right) dx + \left(-\frac{\sqrt{2}F\sqrt{a}\sqrt{b}}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}G\sqrt{a}\sqrt{b}}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{i\sqrt{2}\frac{\partial}{\partial t}F}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{i\sqrt{2}\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}\right) dy$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\,\partial_t + \frac{1}{2}\sqrt{2}\,\partial_z$$

$$\Delta = -\frac{1}{2}\sqrt{2}\,\partial_t - \frac{1}{2}\sqrt{2}\,\partial_z$$

$$\delta = -\frac{\sqrt{2}Fa}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}Ga}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}\,\partial_t + \frac{1}{\sqrt{2}\frac{\partial}{\partial t}F + \sqrt{2}\frac{\partial}{\partial t}G}\,\partial_x - \frac{\sqrt{2}F\sqrt{a}\sqrt{b}}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}G\sqrt{a}\sqrt{b}}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\frac{\partial}{\partial t}F}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}\,\partial_y$$

$$\overline{\delta} = -\frac{\sqrt{2}Fa}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}Ga}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}\,\partial_t + \frac{1}{\sqrt{2}\frac{\partial}{\partial t}F + \sqrt{2}\frac{\partial}{\partial t}G}\,\partial_x - \frac{\sqrt{2}F\sqrt{a}\sqrt{b}}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}G\sqrt{a}\sqrt{b}}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{i\sqrt{2}\frac{\partial}{\partial t}F}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{i\sqrt{2}\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}\,\partial_y$$

Spin Coefficients

$$\rho = \frac{i\sqrt{2}a\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}a\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}$$

$$\sigma = \frac{i\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{i\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}$$

$$\kappa = \frac{1}{4}\sqrt{2}a$$

$$\mu = \frac{i\sqrt{2}a\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}a\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}$$

$$\lambda = \frac{i\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{i\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}$$

$$\nu = -\frac{1}{4}\sqrt{2}a$$

$$\alpha = \frac{\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} - \frac{\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\frac{\partial^2}{\partial t\partial y}F}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\frac{\partial^2}{\partial t\partial y}G}{4\left(\frac{\partial}{\partial t}F + \frac{\partial}{\partial t}G\right)}$$

$$\beta = -\frac{\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\frac{\partial^2}{\partial t\partial y}F}{4\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\frac{\partial^2}{\partial t\partial y}G}{4\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)}$$

$$\tau = \frac{1}{4}\sqrt{2}a$$

$$\pi = -\frac{1}{4}\sqrt{2}a$$

$$\epsilon = -\frac{i\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}F}{8\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}a\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}G}{8\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}a\frac{\partial}{\partial y}G}{8\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)}$$

$$\gamma = -\frac{i\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}F}{8\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}a\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}G}{8\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)} + \frac{i\sqrt{2}a\frac{\partial}{\partial y}G}{8\left(\frac{\partial}{\partial t}F+\frac{\partial}{\partial t}G\right)}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = {} & \frac{a^2\frac{\partial}{\partial t}F^2}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{ab\frac{\partial}{\partial t}F^2}{8\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{a^2\frac{\partial}{\partial y}F^2}{8\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{a^2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} \\
& + \frac{ab\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial y}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{a^2\frac{\partial}{\partial t}G^2}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{ab\frac{\partial}{\partial t}G^2}{8\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} \\
& - \frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{a^2\frac{\partial}{\partial y}F\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}G\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{a^2\frac{\partial}{\partial y}G^2}{8\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{\frac{\partial}{\partial t}F\frac{\partial^3}{(\partial t)^3}F}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} \\
& - \frac{\frac{\partial^3}{(\partial t)^3}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{\frac{\partial}{\partial t}F\frac{\partial^3}{(\partial t)^3}G}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{\frac{\partial}{\partial t}G\frac{\partial^3}{(\partial t)^3}G}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{\frac{\partial}{\partial t}F\frac{\partial}{\partial x}a}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{\frac{\partial}{\partial t}G\frac{\partial}{\partial x}a}{4\left(\frac{\partial}{\partial t}F^2+2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}
\end{aligned}$$

$$
\begin{aligned}
\Phi_{01} = & -\frac{i\,Fa^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} - \frac{i\,Ga^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} - \frac{i\,a^2\frac{\partial}{\partial t}F^2\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& + \frac{i\,ab\frac{\partial}{\partial t}F^2\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} + \frac{i\,Fa^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} + \frac{i\,Ga^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& + \frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} - \frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} - \frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial y)^2}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& + \frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial y)^2}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} + \frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} + \frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& - \frac{i\,Fa^2\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} - \frac{i\,Ga^2\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} + \frac{i\,Fab\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} - \dots
\end{aligned}
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\begin{aligned}
\Phi_{02} = & -\frac{a^2\frac{\partial}{\partial t}F^2}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{ab\frac{\partial}{\partial t}F^2}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{a^2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{ab\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial y}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} \\
& - \frac{a^2\frac{\partial}{\partial t}G^2}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{ab\frac{\partial}{\partial t}G^2}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial}{\partial y}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{i\,a\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} \\
& + \frac{i\,a\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{i\,\sqrt{a}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{i\,a\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{i\,\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} \\
& + \frac{i\,a\frac{\partial}{\partial y}F\frac{\partial^2}{(\partial t)^2}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{i\,a\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} - \frac{i\,a\frac{\partial}{\partial t}G\frac{\partial^2}{\partial t\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{i\,a\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} \\
& + \frac{i\,a\frac{\partial^2}{(\partial t)^2}G\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{\frac{\partial}{\partial t}F\frac{\partial^3}{(\partial t)^3}F}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{\frac{\partial^3}{(\partial t)^3}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \frac{\frac{\partial}{\partial t}F\frac{\partial^3}{(\partial t)^3}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{10} = {} & \frac{i\,Fa^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Ga^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,a^2\frac{\partial}{\partial t}F^2\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& -\frac{i\,ab\frac{\partial}{\partial t}F^2\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Fa^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Ga^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& -\frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial y)^2}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& -\frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial y)^2}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& +\frac{i\,Fa^2\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Ga^2\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Fab\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{11} = {} & -\frac{ab\frac{\partial}{\partial t}F^3}{16\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& -\frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F^2\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& +\frac{Fab\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{Gab\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{3\,a^2\frac{\partial}{\partial t}F\frac{\partial}{\partial y}F^2}{16\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& -\frac{ab\frac{\partial}{\partial t}F\frac{\partial}{\partial y}F^2}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{Fab\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial y)^2}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{Gab\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial y)^2}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)} \\
& +\frac{ab\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G}{16\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\dots
\end{aligned}
$$

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\begin{aligned}
\Phi_{12} = &-\frac{i\,Fa^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Ga^2\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,a^2\frac{\partial}{\partial t}F^2\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}\\
&+\frac{i\,ab\frac{\partial}{\partial t}F^2\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Fa^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Ga^2\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}\\
&+\frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial y}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial y)^2}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}\\
&+\frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial y)^2}F}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Fa^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Ga^{\frac{3}{2}}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}\\
&-\frac{i\,Fa^2\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\frac{i\,Ga^2\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{8\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}+\frac{i\,Fab\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^3+3\,\frac{\partial}{\partial t}F^2\frac{\partial}{\partial t}G+3\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G^2+\frac{\partial}{\partial t}G^3\right)}-\dots
\end{aligned}
$$

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\begin{aligned}
\Phi_{20} = &-\frac{a^2\frac{\partial}{\partial t}F^2}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{ab\frac{\partial}{\partial t}F^2}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}-\frac{a^2\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}-\frac{ab\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial y}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}\\
&-\frac{a^2\frac{\partial}{\partial t}G^2}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{ab\frac{\partial}{\partial t}G^2}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}-\frac{a^{\frac{3}{2}}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial}{\partial y}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{i\,a\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}F}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}\\
&-\frac{i\,a\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}F}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}-\frac{i\,\sqrt{a}\sqrt{b}\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial t}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{i\,a\frac{\partial^2}{\partial t\partial y}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{i\,\sqrt{a}\sqrt{b}\frac{\partial}{\partial t}F\frac{\partial^2}{(\partial t)^2}G}{2\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}\\
&-\frac{i\,a\frac{\partial}{\partial y}F\frac{\partial^2}{(\partial t)^2}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{i\,a\frac{\partial}{\partial t}F\frac{\partial^2}{\partial t\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{i\,a\frac{\partial}{\partial t}G\frac{\partial^2}{\partial t\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}-\frac{i\,a\frac{\partial^2}{(\partial t)^2}F\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}\\
&-\frac{i\,a\frac{\partial^2}{(\partial t)^2}G\frac{\partial}{\partial y}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{\frac{\partial}{\partial t}F\frac{\partial^3}{(\partial t)^3}F}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{\frac{\partial^3}{(\partial t)^3}F\frac{\partial}{\partial t}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\frac{\frac{\partial}{\partial t}F\frac{\partial^3}{(\partial t)^3}G}{4\left(\frac{\partial}{\partial t}F^2+2\,\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G+\frac{\partial}{\partial t}G^2\right)}+\dots
\end{aligned}
$$

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\begin{aligned}
\Phi_{21} = {} & \frac{i\,F a^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,G a^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,a^2 \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial y} F}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& - \frac{i\,ab \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial y} F}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,F a^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,G a^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& - \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& - \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& + \frac{i\,F a^2 \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,G a^2 \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,F ab \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{a^2 \frac{\partial}{\partial t} F^2}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{ab \frac{\partial}{\partial t} F^2}{8\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial}{\partial y} F}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^2 \frac{\partial}{\partial y} F^2}{8\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G}{2\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} \\
& + \frac{ab \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial y} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^2 \frac{\partial}{\partial t} G^2}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{ab \frac{\partial}{\partial t} G^2}{8\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} \\
& - \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial}{\partial y} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^2 \frac{\partial}{\partial y} F \frac{\partial}{\partial y} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} G \frac{\partial}{\partial y} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^2 \frac{\partial}{\partial y} G^2}{8\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} F \frac{\partial^3}{(\partial t)^3} F}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} \\
& - \frac{\frac{\partial^3}{(\partial t)^3} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} F \frac{\partial^3}{(\partial t)^3} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} G \frac{\partial^3}{(\partial t)^3} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} F \frac{\partial}{\partial x} a}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} G \frac{\partial}{\partial x} a}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & -\frac{a^2 \frac{\partial}{\partial t} F^3}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{ab \frac{\partial}{\partial t} F^3}{48\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{F a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& - \frac{G a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& + \frac{G a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{F ab \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{G ab \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& + \frac{a^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial y} F^2}{48\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{ab \frac{\partial}{\partial t} F \frac{\partial}{\partial y} F^2}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{F ab \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& + \frac{G ab \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{a^2 \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{ab \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G}{48\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \dots
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = {} & \frac{a^2 \frac{\partial}{\partial t} F^2}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{ab \frac{\partial}{\partial t} F^2}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G}{2\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{ab \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G}{2\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial y} F \frac{\partial}{\partial t} G}{2\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} \\
& + \frac{a^2 \frac{\partial}{\partial t} G^2}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{ab \frac{\partial}{\partial t} G^2}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial}{\partial y} G}{2\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{i\,a \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} \\
& - \frac{i\,a \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{i\,\sqrt{a}\sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial t} G}{2\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{i\,a \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{i\,\sqrt{a}\sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial t)^2} G}{2\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} \\
& - \frac{i\,a \frac{\partial}{\partial y} F \frac{\partial^2}{(\partial t)^2} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{i\,a \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{i\,a \frac{\partial}{\partial t} G \frac{\partial^2}{\partial t \partial y} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{i\,a \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} \\
& - \frac{i\,a \frac{\partial^2}{(\partial t)^2} G \frac{\partial}{\partial y} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} F \frac{\partial^3}{(\partial t)^3} F}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial^3}{(\partial t)^3} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} F \frac{\partial^3}{(\partial t)^3} G}{4\left(\frac{\partial}{\partial t} F^2 + 2\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \cdots
\end{aligned}
$$

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\begin{aligned}
\Psi_1 = {} & \frac{i\,F a^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,G a^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,a^2 \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial y} F}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& - \frac{i\,ab \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial y} F}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,F a^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,G a^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& - \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& - \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
& + \frac{i\,F a^2 \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,G a^2 \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,F ab \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_2 = &-\frac{a^2 \frac{\partial}{\partial t} F^3}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{ab \frac{\partial}{\partial t} F^3}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{F a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
&+ \frac{G a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{G a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
&- \frac{F ab \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{G ab \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{a^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial y} F^2}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
&+ \frac{ab \frac{\partial}{\partial t} F \frac{\partial}{\partial y} F^2}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{F ab \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{G ab \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
&- \frac{a^2 \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{ab \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G}{12\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{6\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = &-\frac{i\,F a^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,G a^2 \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,a^2 \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial y} F}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
&+ \frac{i\,ab \frac{\partial}{\partial t} F^2 \frac{\partial}{\partial y} F}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,F a^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,G a^2 \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
&+ \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial y} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
&+ \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial y)^2} F}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,F a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,G a^{\frac{3}{2}} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} \\
&- \frac{i\,F a^2 \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \frac{i\,G a^2 \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{8\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} + \frac{i\,F ab \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{4\left(\frac{\partial}{\partial t} F^3 + 3\,\frac{\partial}{\partial t} F^2 \frac{\partial}{\partial t} G + 3\,\frac{\partial}{\partial t} F \frac{\partial}{\partial t} G^2 + \frac{\partial}{\partial t} G^3\right)} - \dots
\end{aligned}
$$

$$\Psi_4 = \frac{a^2 \frac{\partial}{\partial t} F^2}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{ab \frac{\partial}{\partial t} F^2}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G}{2 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{ab \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G}{2 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial y} F \frac{\partial}{\partial t} G}{2 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)}$$
$$+ \frac{a^2 \frac{\partial}{\partial t} G^2}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{ab \frac{\partial}{\partial t} G^2}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{a^{\frac{3}{2}} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial}{\partial y} G}{2 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{i\, a \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} F}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)}$$
$$+ \frac{i\, a \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} F}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{i \sqrt{a} \sqrt{b} \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial t} G}{2 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{i\, a \frac{\partial^2}{\partial t \partial y} F \frac{\partial}{\partial t} G}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{i \sqrt{a} \sqrt{b} \frac{\partial}{\partial t} F \frac{\partial^2}{(\partial t)^2} G}{2 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)}$$
$$+ \frac{i\, a \frac{\partial}{\partial y} F \frac{\partial^2}{(\partial t)^2} G}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{i\, a \frac{\partial}{\partial t} F \frac{\partial^2}{\partial t \partial y} G}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{i\, a \frac{\partial}{\partial t} G \frac{\partial^2}{\partial t \partial y} G}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} + \frac{i\, a \frac{\partial^2}{(\partial t)^2} F \frac{\partial}{\partial y} G}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)}$$
$$+ \frac{i\, a \frac{\partial^2}{(\partial t)^2} G \frac{\partial}{\partial y} G}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} F \frac{\partial^3}{(\partial t)^3} F}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial^3}{(\partial t)^3} F \frac{\partial}{\partial t} G}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \frac{\frac{\partial}{\partial t} F \frac{\partial^3}{(\partial t)^3} G}{4 \left(\frac{\partial}{\partial t} F^2 + 2 \frac{\partial}{\partial t} F \frac{\partial}{\partial t} G + \frac{\partial}{\partial t} G^2\right)} - \cdots$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"